

Paper XII — The Charged-Lepton Shape Angle: $3\delta = Q$, the Koide Phase as the E_g Representation Weight $\frac{2}{9}$, and a Saturation Self-Duality

B. Ludwig *Independent researcher*

*Series: “A Universe in a Bottle” — companion to Paper IX (Lepton Sector). Paper IX derived the Koide **amplitude** ($Q = \frac{2}{3}$, the 45° equipartition); this paper derives the Koide **phase** — the second, hitherto-fitted invariant of the charged-lepton spectrum — and shows the two are the same representation ratio read radially and angularly.*

New (2026-06-30). Consolidates Findings F174–F177 (with F92, F49, F75, F93, F145, F86).

Revision (2026-07-01): re-issued as Paper XII (was Paper XIII) following the deprecation of the horizon-free “dielectric black hole” paper and the consequent renumbering of Papers IX–XIII down by one.

Abstract

The charged-lepton masses are fixed by two invariants of a single E_g order parameter on the second-neighbour shell of the BCC lattice: a **radial** invariant (the Koide ratio Q , set by the order-parameter amplitude) and an **angular** invariant (the Koide–Foot phase δ). Paper X derived the radial one, $Q = \frac{2}{3}$, as a 45° cubic-amplitude equipartition. This paper derives the angular one. We first pin it empirically: the charged leptons require $\delta^* = \frac{2}{9}$ rad ($3\delta^* = \frac{2}{3}$ rad), in agreement with our extraction at 0.9σ and with Brannen’s independent circulant fit $\delta = 0.2222220(19)$. A decisive structural fact follows: the *geometric* (algebraic-cosine) candidates for $\cos 3\delta^*$ are excluded at $10\text{--}31\sigma$, while the *rational-radian* value $3\delta^* = \frac{2}{3}$ survives — so the phase is a representation weight carried as an angle, not a crystallographic angle. We then derive the number from the lattice: $\frac{2}{9} = \dim(E_g)/\dim(T_{1u} \otimes T_{1u})$ is the exact O_h weight of the generation-splitting channel (verified by character projection), the same “2-of-9” invariant that gives the second-shell Weinberg ratio $\sin^2 \theta_W = \frac{2}{9}$. The radial and angular invariants are unified by $Q = \dim(E_g)/\dim(T_{1u}) = \frac{2}{3}$ and $\delta^* = \dim(E_g)/\dim(T_{1u})^2 = \frac{2}{9}$, related by $3\delta^* = Q$ — the factor $\dim(T_{1u}) = 3$ being the threefold of $\cos 3\delta$. We identify the dynamical principle behind $3\delta^* = Q$ as a **saturation self-duality** (angular invariant = radial invariant), and trace its origin to the model’s BPS/saturation structure: the radial half is *derived* (the 45° self-dual pair rotation), while the angular half is shown to be exactly the one shared nonperturbative residual of the confining sector, $C/|B| = 1/(2 \cos \frac{2}{3}) = 0.636$. Using the derived $\delta^* = \frac{2}{9}$ with the derived $\sqrt{2}$ amplitude reproduces the full charged-lepton mass ratios to $\leq 0.007\%$ with **zero shape parameters**. The single honest residual — a first-principles fix of the saturated-condensate ratio, equivalently of the η^2 – δ joint-exactness — is recorded precisely.

1. Introduction

The charged-lepton masses m_e, m_μ, m_τ contain one famous, precise, and standard-model-unexplained relation — the Koide ratio

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.6666605 \approx \frac{2}{3},$$

satisfied to one part in 10^5 . In the Koide–Foot circulant parametrisation

$$\sqrt{m_a} = \mu(1 + \sqrt{2} \cos(\delta + \frac{2\pi a}{3})), \quad a = 0, 1, 2, \quad (1)$$

the spectrum is described by *two* numbers beyond the overall scale μ : the **amplitude** (here $\sqrt{2}$, equivalently $\eta^2 = \frac{1}{2}$), which controls Q , and the **phase** δ , which controls the mass ratios at fixed Q . Paper IX (Finding F80/F92) derived the amplitude: $Q = 1/(3\cos^2\phi)$ and the cubic-amplitude vector sits at $\phi = 45^\circ$, giving $Q = \frac{2}{3}$ — the radial invariant. The phase δ was left as the sector’s one fitted number (Findings F93/F118/F150), entering through the E_g Landau brake $\cos 3\delta^* = -B/2C$.

This paper closes that gap. We show the phase is not free: it is the representation weight of the splitting channel, $\delta^* = \frac{2}{9}$ rad, fixed by the same lattice structure that fixes the amplitude, and the two invariants are unified by the relation $3\delta^* = Q$.

The model background needed is light: the three charged leptons fill the cubic vector irrep T_{1u} of the octahedral point group O_h (Paper VIII, Finding F75); the generation-splitting order parameter is an E_g condensate on the BCC second-neighbour (cube-axis) shell (Paper IX, Finding F93); and the condensate sits at a saturation/BPS point (Findings F73/F82/F86).

2. The phase is $\delta^* = \frac{2}{9}$ — and it is a rational radian, not a geometric angle

2.1 The value

Extracting the phase of (1) from the PDG masses gives $\delta^* = 0.222229$ rad and $3\delta^* = 0.666689$ rad, consistent with the rational $\frac{2}{9}$ (resp. $\frac{2}{3}$) at 0.9σ (the uncertainty 2.5×10^{-5} is dominated by m_τ). The value is corroborated independently by Brannen’s circulant fit, $\delta = 0.2222220(19)$ — i.e. $\frac{2}{9}$ to seven digits — obtained without reference to this model.

2.2 The decisive structural clue

Which exact form is it? Testing candidates for $\cos 3\delta^* = 0.785874(16)$:

candidate	type	value	deviation
$\cos(2/3)$ — i.e. $3\delta^* = \frac{2}{3}$ rad	rational radian	0.785887	0.9σ
11/14	algebraic (rational cosine)	0.785714	10σ — excluded
$\pi/4$	“geometric” angle	0.785398	31σ — excluded
$1/\sqrt{2}$	algebraic (the $m_e=0$ limit, §4)	0.707107	far

The *geometric* / algebraic-cosine forms are excluded; only the **rational-radian** $3\delta^* = \frac{2}{3}$ survives. This is the central clue. A crystallographic angle or a group-theoretic projection yields an algebraic cosine (a rational multiple of π , or a root of a polynomial); a rational *number of radians* is the signature of a **representation weight** or topological moment carried as a phase, not a geometric

angle. (It also explains why the Landau brake coefficient $\lambda_6 \propto 1/\cos \frac{2}{3}$ is *not* a clean rational, Finding F164: the simplicity lives in the angle, not the coefficient.)

3. Deriving $\frac{2}{9}$ from the lattice: the E_g representation weight

The generation order parameter is a Hermitian bilinear of the T_{1u} triplet. Under O_h ,

$$T_{1u} \otimes T_{1u} = A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}, \quad \dim = 1 + 2 + 3 + 3 = 9. \quad (2)$$

The unique channel that splits the three generations *without mixing axes* is E_g (Finding F93). Its weight in the nine-dimensional bilinear is

$$\boxed{\frac{\dim(E_g)}{\dim(T_{1u} \otimes T_{1u})} = \frac{2}{9}} \quad (3)$$

verified by explicit O_h character projection over all 24 rotations (the multiplicity of E in $T_{1u} \otimes T_{1u}$ is exactly 1). Equation (3) is the lattice origin of the number: pure second-shell representation theory, no dynamics, no fit.

The same $\frac{2}{9}$ appears in the **gauge** sector on the **same** shell: the Weinberg ratio $\sin^2 \theta_W = \frac{2}{9} = 2 \text{ sublattices} / (2 + 7 \text{ bond axes})$ (Finding F49) is the identical “2 special / 9 total” structure. So $\frac{2}{9}$ is a genuine invariant of the BCC second shell, appearing in two independent sectors.

Why a representation weight, not a winding. Under the threefold C_3 (a 120° lattice rotation), the E_g doublet carries character $-1 = 2 \cos \frac{2\pi}{3}$: it advances by a $\frac{2\pi}{3}$ phase, giving the $\cos 3\delta$ Landau invariant. A literal *geometric* winding would therefore be a fraction of 2π — an algebraic cosine — which §2.2 excluded. Hence the rational-radian $\frac{2}{9}$ is the representation weight carried as the phase, exactly as §2.2 demanded, and as the external “topological moment” derivation of $\delta = \frac{2}{9}$ (ZIP) realises in three-dimensional configuration space.

4. The unification $3\delta = Q$ and the self-duality principle

Two facts combine. First, the radial invariant has an exact representation form,

$$Q = \frac{\dim(E_g)}{\dim(T_{1u})} = \frac{2}{3} \quad (4)$$

(Paper IX’s 45° equipartition in representation language). Second, the angular invariant is 3δ , the 3 being $\dim(T_{1u})$ — the same threefold as $\cos 3\delta$ (§3). The **dynamical principle** is that at saturation the angular and radial invariants coincide,

$$\boxed{3\delta^* = Q} \quad (\text{saturation self-duality}), \quad (5)$$

which, with (4) and the threefold, gives

$$\delta^* = \frac{\dim(E_g)}{\dim(T_{1u})^2} = \frac{2}{9}. \quad (6)$$

Equation (5) is the angular twin of Paper IX’s radial equipartition; the factor $\dim(T_{1u}) = 3$ converts the radial ratio $\frac{2}{3}$ into the angular weight $\frac{2}{9}$.

The principle is a genuine condition, not an identity. Self-duality $3\delta = Q$ holds for the physical spectrum to 2.8×10^{-5} , but *fails* away from it: sending $m_e \rightarrow 0$ (where, in the idealised two-mass texture, $3\delta \rightarrow \pi/4$ exactly, Finding F96) gives $3\delta \neq Q$. Thus (5) *selects* the physical lightest mass; it is not automatic. A naive maximum-entropy/equipartition over the masses does not reproduce it (it returns the symmetric points $\delta = 0, \pi/3$) — the working principle is specifically the radial=angular self-duality.

5. Origin of the self-duality: the BPS/saturation structure

5.1 The radial half is derived

The radial self-duality follows from the model’s saturation (BPS) point. The bound-pair rotation peaks at $\phi = 45^\circ$ (Finding F82, coupling-independent), the **self-dual** point where $\sin \phi = \cos \phi$ — the Bogomolny condition. There the pair amplitude $y = \sqrt{2} \sin \phi = 1$ saturates the wall (Findings F73/F101), giving $\eta^2 = \frac{1}{2}$, hence $Q = \frac{2}{3}$, and equivalently $\angle(\sqrt{\mathbf{m}}, (1, 1, 1)) = 45^\circ$ (Foot). The radial invariant is therefore a *derived* BPS self-duality — “ 45° everywhere.”

5.2 The angular half is the one shared residual

The angular self-duality does not close by the standard route. For the E_g field $E = \int [\frac{1}{2}\delta'^2 + W(\delta)]$, the Bogomolny first-order condition $\delta' = \sqrt{2W}$ fixes the domain-wall tension, not the vacuum angle, which remains the brake minimiser $\cos 3\delta^* = -B/2C$. Nor is δ^* fixed by a *second* geometric self-duality: $\sqrt{\mathbf{m}}$ is not at 45° to any natural second reference (the closest, the $3z^2 - r^2$ axis, is 1.4° off). Instead, (5) is *exactly equivalent* to the induced brake ratio taking a specific value:

$$3\delta^* = Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{C}{|B|} = \frac{1}{2 \cos \frac{2}{3}} = 0.636. \quad (7)$$

This is the same single nonperturbative IR residual that governs the confining sector — the saturated-condensate solve that also fixes the chiral-symmetry-breaking point, $\sqrt{\sigma}/f_\pi$, and the Λ scheme constant (Findings F145/F124/F144/F150, summarised in F172). (It is near $2/\pi = 0.6366$ but not equal, so not a clean closed form.) The angular self-duality is thus a *physical restatement* of that one residual, giving it a sharp target ($C/|B| = 0.636$) and a clean meaning: it is the number that makes the angular invariant equal the BPS-derived radial invariant.

6. Result: the charged-lepton spectrum with zero shape parameters

Using the **derived** phase $\delta^* = \frac{2}{9}$ (3) and the **derived** amplitude $\eta^2 = \frac{1}{2}$ (Paper IX) in (1) — no fitted shape parameter, only the overall scale μ :

ratio	model ($\delta^* = \frac{2}{9}, \sqrt{2}$)	PDG	deviation
m_μ/m_e	206.770	206.7683	+0.001%
m_τ/m_e	3477.47	3477.228	+0.007%

The entire charged-lepton *shape* follows from two representation numbers, $\dim(E_g)/\dim(T_{1u}) = \frac{2}{3}$ and $\dim(E_g)/\dim(T_{1u})^2 = \frac{2}{9}$, plus one scale. The residual 0.007% is the $\sim 0.9\sigma$ amplitude–phase tension (§7) at current mass precision.

7. Honest residuals

1. **Joint exactness.** A free fit gives $\eta^2 = 0.49999$ and $\delta = 0.222229$ — each within $\sim 1\sigma$ of $\frac{1}{2}$ and $\frac{2}{9}$, but the data do not enforce both *exactly* at present precision (Brannen flags the potential conflict; it may be mass-scheme dependent). $\delta^* = \frac{2}{9}$ is a corroborated, lattice-derived value, validated three ways (rational-radian exclusion of geometric alternatives; the $\leq 0.007\%$ spectrum; the representation identity), but the joint $\{\eta^2 = \frac{1}{2}, \delta = \frac{2}{9}\}$ exactness awaits more precise masses.
2. **The one nonperturbative number.** What is *not* derived from first principles is the saturated-condensate ratio (7), $C/|B| = 1/(2 \cos \frac{2}{3}) = 0.636$ — equivalently, the dynamical origin of the angular self-duality. This is the single residual shared across the confining sector (Finding F172); this paper gives it its sharpest statement: *compute $C/|B|$; the self-dual prediction is 0.636.*

Neither residual is a free shape parameter of the lepton spectrum: both reduce to one confining-sector number with a definite target.

8. Conclusion

The charged-lepton spectrum is fixed by one E_g order parameter whose two invariants are the same second-shell representation ratio read two ways: radially, $Q = \dim(E_g)/\dim(T_{1u}) = \frac{2}{3}$ (Paper IX), and angularly, $\delta^* = \dim(E_g)/\dim(T_{1u})^2 = \frac{2}{9}$ (this paper), unified by the saturation self-duality $3\delta^* = Q$. The phase $\frac{2}{9}$ is the exact O_h weight of the generation-splitting channel — the same “2-of-9” that gives the second-shell Weinberg ratio — and, being a rational radian rather than a geometric angle, it is a representation weight carried as a phase. The radial half of the self-duality is derived from the model’s 45° BPS saturation; the angular half is the one nonperturbative residual of the confining sector, now sharply targeted. With both invariants derived, the full charged-lepton shape is reproduced to $\leq 0.007\%$ with no fitted shape parameter, reducing the lepton-mass problem to a single overall scale and one shared confining-sector number.

Findings and modules

- **Findings:** F174 (the angle pinned to $\frac{2}{9}$; rational-radian exclusion), F175 (the E_g representation weight $\dim E_g/\dim(T_{1u} \otimes T_{1u}) = \frac{2}{9}$; zero-shape-parameter spectrum), F176 (the self-duality principle $3\delta = Q$), F177 (BPS completion: radial derived, angular = the shared

residual). Supporting: F92/F80 (Koide amplitude / 45°), F75 (three generations / T_{1u}), F93 (the E_g second-shell condensate), F49 (the gauge $\frac{2}{9}$), F86/F73/F82/F101 (BPS / saturation), F145/F124/F144/F150/F172 (the shared IR residual), F96 (the $m_e = 0$ anchor), F164 (the non-rational λ_6).

- **Verification scripts:** `tests/findings/test_F174_shape_angle_2_9.py`, `test_F175_lattice_2_9_eg_w`, `test_F176_saturation_self_duality.py`, `test_F177_bps_self_duality_completion.py` (all PASS; residuals as quoted).

External references

- Y. Koide, *Lett. Nuovo Cimento* **34** (1982) 201; *Phys. Lett. B* **120** (1983) 161 — the lepton mass relation.
- R. Foot, arXiv:hep-ph/9402242 — the geometric (45°) interpretation.
- C. A. Brannen, *The Lepton Masses* (brannenworks.com/MASSES2.pdf) — the circulant fit $\eta^2 = 0.500003(23)$, $\delta = 0.2222220(19)$.
- “Derivation of the Koide Formula from the Zero-Interaction Principle” — $\delta = \frac{2}{9}$ as a difference of topological moments in 3D configuration space.
- A. Deur, S. J. Brodsky, C. D. Roberts, *Prog. Part. Nucl. Phys.* (rev.) and arXiv:1801.10164, 1912.08232 — the process-independent IR fixed point (context for the nonperturbative residual of §5.2).
- E. B. Bogomolny, *Sov. J. Nucl. Phys.* **24** (1976) 449 — the BPS/self-dual bound.